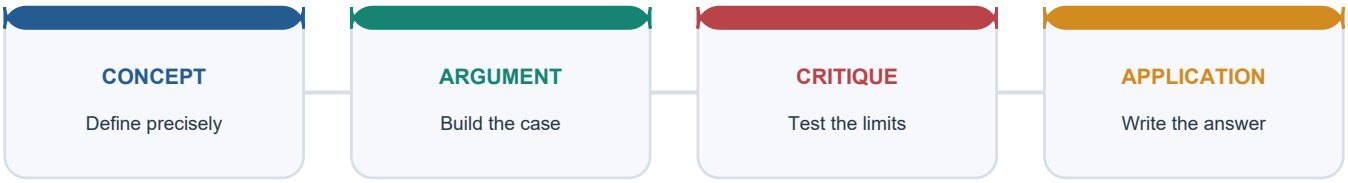
COMPLETE LEARNING SESSION

Tsunami and Coastal Hazard Management - Complete Learning Session

GS-III Disaster Management | Concepts, evidence, practice and answer writing

LEARNING PATH



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Tsunami and Coastal Hazard Management - Complete Learning Session

Subject: Disaster Management | **GS Paper:** GS-III

This package explains tsunami generation, coastal transformation, India's warning architecture and the last-mile governance needed to convert an advisory into safe evacuation and recovery.

BASIC LEARNING SESSION

Session 1: Definition and coastal-hazard firewall

```
TSUNAMI -> sudden displacement of a large water volume
TIDE -> gravitational cycle of moon and sun
STORM SURGE -> cyclone wind and pressure

Different cause -> different monitoring -> different warning -> different response
```

Plain definition: A tsunami is a series of very long waves generated by sudden displacement of a large water volume.

Technical definition: Tsunami generation is geological or mass-displacement driven and must be separated from astronomical tides and meteorological storm surge.

Exam-ready opening: A high-scoring tsunami answer begins by separating geological displacement from astronomical tides and cyclone-driven storm surge.

Core explanation

- A tsunami is a wave train; the first arrival need not be the largest.
- Deep-ocean wave height may be small even when coastal impact becomes severe.
- Danger continues until an authorised all-clear because multiple arrivals and return flow may follow.

Named evidence in an answer

The Japanese expression 'harbour wave' is a useful named definition, but mechanism matters more than etymology. Sudden water displacement distinguishes a tsunami from a tide and storm surge. This distinction determines the correct monitoring chain: seismic and sea-level observations rather than tide tables or cyclone-track forecasts.

UPSC trap: Never use tsunami, high tide and storm surge as interchangeable terms.

Six answer keywords: tsunami | harbour wave | water displacement | wave train | storm surge | all-clear

Recall line: Displacement wave, not tide or surge; multiple arrivals mean wait for the authorised all-clear.

Session 2: Trigger mechanisms and tsunamigenic conditions

```
EARTHQUAKE / LANDSLIDE / VOLCANIC COLLAPSE / METEORITE IMPACT
                        |
              RAPID WATER DISPLACEMENT
                        |
              TSUNAMI POTENTIAL

Earthquake alone is insufficient:
vertical motion + shallow source + rupture area + displacement matter
```

Plain definition: Earthquakes are the commonest trigger, but any rapid movement that displaces enough water can generate a tsunami.

Technical definition: Tsunamigenic potential depends on source mechanism, depth, rupture geometry, vertical seabed movement and displaced-water volume rather than magnitude alone.

Exam-ready opening: Tsunami generation is a conditional chain: a disturbance becomes dangerous only when it rapidly displaces enough water.

Core explanation

- Subduction megathrusts can produce large vertical seabed displacement.
- Strike-slip, deep or small earthquakes may generate little tsunami energy.
- Submarine landslides are regionally relevant where unstable sediment accumulates.

Named evidence in an answer

The Andaman-Sumatra subduction zone is the central Indian example of a megathrust source. The Basic source also links Ganges and Indus sediment deposition to plausible submarine-landslide scenarios in the Bay of Bengal and Arabian Sea. These examples show why earthquake-only warning logic is incomplete.

UPSC trap: Do not assume every undersea earthquake produces a destructive tsunami.

Six answer keywords: subduction megathrust | vertical displacement | rupture area | submarine landslide | volcanic collapse | tsunamigenic

Recall line: Earthquake is commonest, but vertical displacement and water volume - plus landslide, volcano or impact - determine generation.

Session 3: Deep-ocean propagation

```
SOURCE DISTURBANCE
      |
LONG WAVELENGTH / LONG PERIOD
      |
DEEP WATER -> fast travel, low amplitude
      |
BATHYMETRIC REFRACTION + SOURCE DIRECTIVITY
      |
UNEQUAL ARRIVAL AND ENERGY AT DIFFERENT COASTS
```

Plain definition: In deep water, tsunami energy travels rapidly through the water column while the surface wave may remain visually inconspicuous.

Technical definition: Because wavelength is very large relative to ocean depth, tsunamis behave as shallow-water waves whose speed depends mainly on water depth.

Exam-ready opening: The offshore paradox is exam-useful: a low-amplitude wave can travel rapidly across deep ocean while retaining destructive coastal potential.

Core explanation

- Deeper water generally permits faster propagation.
- Bathymetry refracts wave paths and source directivity creates unequal regional impact.
- Distant sources may provide warning time; near-source coasts may have only minutes.

Named evidence in an answer

A ship in deep water may not notice a destructive tsunami because offshore amplitude can be small while energy remains distributed through the water column. The analytical value of this fact is operational: visual observation is not a substitute for instruments, and arrival time varies with depth and source-coast geometry.

UPSC trap: Low offshore wave height does not imply low coastal risk.

Six answer keywords: long wavelength | long period | deep-ocean propagation | water depth | refraction | source directivity

Recall line: Deep ocean: fast and visually small; depth and directivity control arrival and energy.

Session 4: Shoaling, run-up, inundation and return flow

```
DEEP WATER -> SHALLOW WATER
speed falls -> wavelength shortens -> height/current rise
                        |
                    SHOALING
                        |
RUN-UP (vertical reach) + INUNDATION (horizontal flooding)
                        |
RETURN FLOW -> debris, scour and seaward drag
```

Plain definition: Coastal damage intensifies when a long wave enters shallow water, slows and steepens.

Technical definition: Shoaling compresses tsunami energy near shore, producing site-specific run-up, inundation depth, current and destructive return flow.

Exam-ready opening: Most coastal destruction is explained not at the source but at the shallow-water transition from shoaling to run-up, inundation and return flow.

Core explanation

- Run-up is maximum vertical reach above a reference level.
- Inundation is horizontal inland flooding.
- Withdrawal and return flow can carry people, vehicles, debris and eroded material seaward.

Named evidence in an answer

The formation-to-consequence chain is best expressed as deep-water travel followed by shallow-water shoaling, run-up, inundation and return flow. These terms prevent a vague 'large wave hits coast' answer. They also explain why warning must continue after the first arrival.

UPSC trap: Run-up, offshore wave height and inland inundation distance are not the same measure.

Six answer keywords: shoaling | run-up | inundation | return flow | debris load | scour

Recall line: Shoaling -> run-up -> inundation -> return flow; keep all four terms distinct.

Session 5: Coastal amplification and the risk system

```
OFFSHORE HAZARD
      x
COASTAL GEOMETRY / BATHYMETRY
      x
EXPOSURE (people, ports, utilities)
      x
VULNERABILITY (low elevation, weak routes, degraded buffers)
      /
CAPACITY (warning, shelter, drills)
      =
TSUNAMI DISASTER RISK
```

Plain definition: The same offshore wave can produce very different local consequences.

Technical definition: Tsunami risk emerges from source intensity and coastal transformation interacting with exposure, vulnerability and response capacity.

Exam-ready opening: Tsunami disaster risk is coast-specific because offshore energy is filtered through bathymetry, exposure, vulnerability and response capacity.

Core explanation

- Funnel-shaped bays and estuaries may concentrate energy.
- Low-lying coasts permit greater inland penetration.
- Dunes, mangroves, reefs, cliffs and local topography modify impact but do not guarantee safety.

Named evidence in an answer

The 2004 pattern of heavier damage in low-lying, densely populated areas is the named evidence for this risk-system view. Heavy damage where sand dunes had been mined additionally shows how development choices alter vulnerability. Therefore no universal safe distance or height can replace local mapping.

UPSC trap: Do not infer local impact from earthquake magnitude alone.

Six answer keywords: bathymetry | coastal geometry | exposure | vulnerability | capacity | site-specific risk

Recall line: Local risk = hazard transformation x exposure x vulnerability, reduced by maintained capacity.

Session 6: India's source geography and the 2004 Indian Ocean tsunami

```
EASTERN SOURCE: ANDAMAN-SUMATRA SUBDUCTION ZONE
  -> Andaman and Nicobar near-source exposure
  -> Bay of Bengal / east-coast exposure

WESTERN SOURCE: MAKRAN SUBDUCTION ZONE
  -> Arabian Sea / west-coast exposure

26 DECEMBER 2004 -> institutional turning point -> Indian warning system
```

Plain definition: India faces both eastern and western tsunami sources, with different arrival times and coastal consequences.

Technical definition: Indian hazard geography includes the Andaman-Sumatra and Makran subduction systems plus local landslide or volcanic sources.

Exam-ready opening: India's preparedness must cover both the Andaman-Sumatra and Makran source systems, with 2004 as the central institutional lesson.

Core explanation

- The 26 December 2004 event originated in the Sumatra-Andaman megathrust region.
- Andaman and Nicobar, Tamil Nadu, Puducherry, Andhra Pradesh, Kerala and other regional coasts were affected.
- The disaster triggered major post-event investment in India's warning architecture.

Named evidence in an answer

Use 2004 for mechanisms and lessons rather than unsourced casualty recital. Natural barriers such as mangroves, forests, dunes and cliffs reduced damage under particular site conditions, while mined dunes were associated with heavier losses. This supports mixed protection and risk-sensitive development.

UPSC trap: Both coasts are vulnerable, but source, arrival time and inundation are not uniform.

Six answer keywords: Andaman-Sumatra | Makran | 26 December 2004 | near-source exposure | natural buffers | institution-building

Recall line: Andaman-Sumatra east, Makran west; use 26 December 2004 for lessons, not casualty recital.

Session 7: ITEWC and the multi-sensor warning architecture

```
SEISMIC NETWORK -> locate and characterise earthquake
         |
NUMERICAL MODEL -> estimate travel and threat
         |
BOTTOM PRESSURE RECORDERS -> open-ocean confirmation
         |
TIDE GAUGES -> coastal sea-level observation
         |
ITEWC / INCOIS -> staged regional advisory
```

Plain definition: India's warning system combines seismic detection, modelling and sea-level confirmation.

Technical definition: The Indian Tsunami Early Warning Centre at INCOIS is a 24x7 UNESCO-IOC Tsunami Service Provider under IOTWMS, operational since 15 October 2007.

Exam-ready opening: India's warning strength lies in a multi-sensor chain at ITEWC, not in earthquake detection alone.

Core explanation

- Official current anchors distinguish roughly ten-minute earthquake detection from roughly twenty-minute advisory dissemination.
- BPRs and tide gauges test whether a seismic event generated a propagating wave.
- The system uses national and international data and serves India within a regional framework.

Named evidence in an answer

ITEWC at INCOIS, Hyderabad, is the institutional anchor. Its multi-sensor design prevents an earthquake solution from being mistaken for confirmed tsunami impact. The ten-minute and twenty-minute figures describe different stages and should never be collapsed into one warning-time claim.

UPSC trap: Detection capability does not itself prove that a last-mile warning reached or moved every exposed person.

Six answer keywords: ITEWC | INCOIS | IOTWMS | BPR | tide gauge | staged advisory

Recall line: ITEWC at INCOIS since 15 October 2007: about 10-minute detection, about 20-minute advisory, with BPRs and tide gauges.

Session 8: Detection, confirmation, advisory and protective authority

```
EARTHQUAKE DETECTED
 -> source parameters estimated
 -> model-based preliminary threat
 -> sea-level observation updates confidence
 -> ITEWC bulletin / advisory
 -> State-district-local protective decision
 -> evacuation / closure / shelter
 -> continued monitoring
 -> authorised local all-clear
```

Plain definition: Scientific assessment and public protective action are linked but institutionally distinct stages.

Technical definition: ITEWC issues technical threat information; competent state, district and local authorities translate it into evacuation, closure and all-clear decisions.

Exam-ready opening: The decisive governance distinction is between ITEWC's technical advisory and the competent local authority's protective decision.

Core explanation

- Initial bulletins may be precautionary because confirmation follows detection.
- Updates may upgrade, downgrade or cancel the threat as observations arrive.
- A cautious early bulletin that later proves unnecessary is not automatically a warning-system failure.

Named evidence in an answer

The central evidence boundary is authority: INCOIS evaluates and advises, while local authorities order evacuation and declare safety. This division prevents both scientific delay and unaccountable local improvisation. Public communication should explain updates so uncertainty does not destroy trust.

UPSC trap: An ITEWC advisory is not itself the local evacuation order or all-clear.

Six answer keywords: preliminary bulletin | confirmation | threat update | protective action | local authority | all-clear

Recall line: ITEWC informs; State-district-local authorities evacuate and issue the all-clear.

Session 9: Inundation modelling, bathymetry and TUNAMI-N2

```
SOURCE SCENARIO
 + deep-ocean bathymetry
 + near-shore bathymetry
 + coastal topography
 + friction / land cover / structures
          |
          v
TRAVEL TIME + RUN-UP + DEPTH + VELOCITY + INUNDATION MAP
          |
EVACUATION / SITING / DESIGN DECISIONS
```

Plain definition: Models convert source and terrain data into planning scenarios for coastal impact.

Technical definition: TUNAMI-N2, customised by ICMAM and INCOIS, supports surge and inundation assessment using bathymetric and topographic inputs.

Exam-ready opening: Inundation models are planning tools whose value depends on high-resolution bathymetry, topography and transparent uncertainty.

Core explanation

- The Indian Naval Hydrographic Department supplies bathymetry and supported re-charting after 2004.
- High-resolution near-shore data remain essential for local accuracy.
- A model line is a scenario boundary with uncertainty, not a guaranteed maximum wave limit.

Named evidence in an answer

The seven survey ships deployed by the Naval Hydrographic Department after 2004 provide named evidence of bathymetry's operational value: sea lanes had to be reopened and seabed information updated. Modelling links that data to evacuation, settlement avoidance and structural design.

UPSC trap: Do not present model output as an exact prediction for every future event.

Six answer keywords: TUNAMI-N2 | ICMAM | bathymetry | topography | inundation model | uncertainty band

Recall line: TUNAMI-N2 + bathymetry + topography -> scenario map, never an exact future boundary.

Session 10: Natural signs and horizontal or vertical evacuation

```
STRONG / LONG SHAKING OR UNUSUAL SEA CHANGE
              |
        IMMEDIATE SELF-EVACUATION
              |
   +----------+----------+
   |                     |
HORIZONTAL            VERTICAL
inland / higher land   engineered designated refuge
   |                     |
route time < arrival time; accessible and maintained
              |
        WAIT FOR ALL-CLEAR
```

Plain definition: Near-source communities may need to evacuate on natural signs before an official message arrives.

Technical definition: Evacuation design compares likely arrival time with accessible movement to safe horizontal ground or an officially designated vertical refuge.

Exam-ready opening: For a near-source tsunami, natural signs and immediate movement to safe ground may save more time than waiting for a formal message.

Core explanation

- Natural signs include strong or long shaking, unusual sea withdrawal or rise and a loud ocean roar.
- Horizontal evacuation is preferred where terrain and time permit.
- Vertical structures require earthquake, hydrodynamic, debris, scour, capacity and accessibility design.

Named evidence in an answer

The critical named instruction is simple: after strong coastal shaking or abnormal sea behaviour, move away from the coast rather than approaching it to observe. An ordinary tall building is not automatically a safe vertical refuge. Routes and structures must be mapped, signed, drilled and officially designated.

UPSC trap: Sea withdrawal is not necessary in every tsunami; absence of one sign does not prove safety.

Six answer keywords: natural warning signs | self-evacuation | horizontal evacuation | vertical refuge | accessible route | official designation

Recall line: Strong shaking or abnormal sea change -> self-evacuate; use designated horizontal or vertical routes.

Session 11: Tsunami Ready and readiness decay

```
RARE HAZARD
 -> memory fades / staff change / signage degrades / exposure changes
 -> drills become ritual
 -> readiness decays

COUNTER-CYCLE:
map -> educate -> warn -> drill -> evaluate indicators -> renew recognition
```

Plain definition: Low frequency makes tsunami preparedness difficult to maintain between events.

Technical definition: UNESCO-IOC Tsunami Ready is a renewable, community-level indicator programme designed to institutionalise warning reception, mapping, outreach and drills.

Exam-ready opening: Tsunami Ready converts a rare-hazard problem into a renewable community-preparedness cycle rather than a one-time badge.

Core explanation

- Venkatraipur in Ganjam and Noliasahi in Jagatsinghpur were India's first recognised communities on 7 August 2020.
- Recognition applies to named communities, not an entire state or country.
- Renewal and repeated drills matter because settlements, tourists, staff and routes change.

Named evidence in an answer

Venkatraipur and Noliasahi are the correct Indian evidence for community-level readiness. Their significance is not that India became nationally 'Tsunami Ready', but that preparedness can be assessed against local indicators. The renewable design directly addresses readiness decay.

UPSC trap: A recognition badge is not permanent proof that equipment, routes and public knowledge remain functional.

Six answer keywords: Tsunami Ready | Venkatraipur | Noliasahi | community recognition | renewal | readiness decay

Recall line: Venkatraipur and Noliasahi, 7 August 2020: renewable community recognition, not national certification.

Session 12: Coastal regulation and layered protection

```
HAZARD / INUNDATION MAP
       |
CRZ-I to CRZ-IV + LOCAL LAND-USE PLAN
       |
   +---+--------------------+
   |                        |
ENGINEERED               ECOSYSTEM
walls / breakwaters      dunes / mangroves / reefs
shelters / retrofits     forests / wetlands / bio-shields
   |                        |
   +------ LOCATION-SPECIFIC PORTFOLIO ------+
```

Plain definition: Coastal safety requires risk-sensitive siting plus complementary engineered and ecosystem measures.

Technical definition: CRZ Notification 2019 category-specific controls, hazard maps and MHA's techno-legal framework guide zoning, design and construction without imposing one universal setback.

Exam-ready opening: Coastal safety requires CRZ-linked siting plus a location-specific mix of engineered and ecosystem protection.

Core explanation

- Engineered measures may face overtopping, scour, maintenance cost and transferred erosion.
- Natural buffers provide friction, elevation and ecological co-benefits but cannot eliminate extreme risk.
- Dune mining and mangrove loss can convert development pressure into disaster vulnerability.

Named evidence in an answer

The 2004 observation of heavier damage where dunes were mined is direct evidence for ecosystem protection. Pair it with sea walls, breakwaters, shelters or retrofitting to show that the appropriate strategy is layered. CRZ categories regulate broader coastal development; tsunami safety is one application.

UPSC trap: Neither a sea wall nor a mangrove belt is a universal standalone solution.

Six answer keywords: CRZ 2019 | hazard zoning | techno-legal framework | engineered protection | ecosystem buffer | layered mitigation

Recall line: CRZ-I to CRZ-IV + maps + engineered and ecosystem layers; no universal setback or barrier.

Session 13: Ports, fisheries, tourism and inclusive warning

```
COASTAL WARNING
  |
  +-- fishers at sea / landing centres / markets
  +-- ports / hazardous cargo / channels / cranes
  +-- tourists / hotels / unfamiliar languages and routes
  +-- islands / limited medical and supply redundancy
  +-- children / older persons / disability / patients / migrants
  |
  v
DIFFERENT MESSAGE + ASSISTANCE + CONTINUITY PLAN
```

Plain definition: A universal bulletin must be translated into sector-specific and accessible action.

Technical definition: Inclusive coastal preparedness differentiates warning channels, evacuation support and continuity planning for livelihoods, critical facilities and populations with unequal mobility or local knowledge.

Exam-ready opening: Last-mile safety is unequal unless warnings and continuity plans are adapted for fishers, ports, tourists, islands and assisted evacuation.

Core explanation

- Fishers require communication at sea and restoration of boats, gear, landing centres and markets.
- Ports require shutdown, hazardous-material control, navigation assessment and reopening plans.
- Tourists need multilingual signage and routes because they may not recognise natural warnings.

Named evidence in an answer

The coast-resource PYQ makes this livelihood link especially useful: ports, fisheries and tourism create value and exposure simultaneously. Accessible audio, visual and text warnings, assisted evacuation and continuity plans turn a technical advisory into equitable protection.

UPSC trap: Do not design preparedness only for permanent, able-bodied residents already familiar with the coast.

Six answer keywords: fisher livelihoods | port continuity | tourist warning | multilingual message | assisted evacuation | island redundancy

Recall line: Different coastal users need different messages, assistance and continuity arrangements.

Session 14: Response, recovery and Build Back Better

```
EVACUATE / ACCOUNT / SEARCH AND RESCUE
        |
HEALTH + WATER + SANITATION + SHELTER + PROTECTION
        |
DEBRIS / SALINITY / PORT / UTILITY RESTORATION
        |
LIVELIHOOD RECOVERY + SAFE HOUSING SITING
        |
UPDATE MAPS / ROUTES / DESIGN / DRILLS
        |
BUILD BACK BETTER WITHOUT BREAKING COASTAL RIGHTS
```

Plain definition: Recovery should reduce future exposure while restoring services, livelihoods and community rights.

Technical definition: Build Back Better integrates damage assessment, safer siting, resilient infrastructure, ecosystem restoration and livelihood-sensitive reconstruction.

Exam-ready opening: Recovery becomes resilience only when Build Back Better restores services and livelihoods without recreating coastal exposure.

Core explanation

- Community, SHGs, NGOs and CBOs support response alongside formal agencies.
- Immediate needs include search-rescue equipment, boats, medical care, water, sanitation and shelter.
- Forced relocation disconnected from fisheries or sea access can create a second livelihood disaster.

Named evidence in an answer

The Naval Hydrographic Department's seven-ship 2004 deployment is a strong response example because it combined sea-lane reopening, aid and bathymetric re-charting. Recovery should then feed lessons back into maps, evacuation routes, critical-facility design and ecosystem restoration.

UPSC trap: Build Back Better does not mean rebuilding the same exposed settlement or relocating people without livelihood access.

Six answer keywords: search and rescue | service restoration | livelihood recovery | safe siting | ecosystem restoration | Build Back Better

Recall line: Response restores life and services; Build Back Better updates siting, routes, design and livelihoods.

Session 15: Integrated tsunami-resilience framework

```
SOURCE -> PROPAGATION -> SHOALING -> LOCAL IMPACT
  |
DETECTION -> CONFIRMATION -> ADVISORY -> LOCAL ACTION
  |
MAP / CRZ -> LAYERED PROTECTION -> EVACUATION
  |
PORT / LIVELIHOOD / INCLUSION -> RESPONSE -> RECOVERY
  |
DRILLS + MAINTENANCE + UPDATED DATA

TRUE RESILIENCE = CONTINUOUS CHAIN, NOT ONE TECHNOLOGY
```

Plain definition: Tsunami resilience depends on a maintained chain from science to safe local action and recovery.

Technical definition: An exam-safe framework integrates source physics, multi-sensor warning, risk-sensitive land use, layered protection, inclusive evacuation, continuity and feedback-based recovery.

Exam-ready opening: A complete tsunami answer must connect source physics, ITEWC, local action, land-use discipline, inclusive evacuation and maintained readiness.

Core explanation

- Tsunamis are low-frequency, high-consequence hazards vulnerable to institutional forgetting.
- Climate change does not generate tsunamis, though sea-level rise and coastal erosion can alter the exposure baseline.

- Compound risk may include shaking, liquefaction, port fire, hazardous spills, high tide and damaged evacuation routes.

Named evidence in an answer

The strongest synthesis uses the 2004 event as the institutional turning point, ITEWC as the technical backbone and Tsunami Ready as the last-mile maintenance instrument. It then applies the mandate-outcome firewall: a fast advisory is an input, while safe evacuation and reduced loss require local evidence.

UPSC trap: Do not label tsunami a climate-induced hazard or equate warning capability with avoided loss.

Six answer keywords: low-frequency hazard | compound risk | ITEWC | Tsunami Ready | last-mile action | continuous readiness

Recall line: 2004 + ITEWC + Tsunami Ready form the evidence spine; the outcome test is safe last-mile action.

BASIC MCQS / REMEDIATION

Q1. Which statement best captures the defining characteristic of a tsunami?

A. A tsunami is a series of large waves of extremely long wavelength and period, usually generated by an undersea disturbance, most commonly earthquake-driven vertical seabed movement. B. A tsunami is any unusually large wave produced by strong onshore winds during a tropical cyclone. C. A tsunami is the periodic rise and fall of sea level caused by gravitational forces of the moon and sun. D. A tsunami is a permanent rise in mean sea level caused by long-term glacial melt.

Answer: A. Explanation: VisionIAS defines a tsunami as long-wavelength, long-period waves from an undersea disturbance, most often vertical seabed movement in an earthquake. The remaining options wrongly invoke storm-surge dynamics, astronomical tides, or long-term sea-level rise, none of which matches this definition.

Q2. Which set lists recognised non-earthquake triggers of tsunami generation?

A. Only changes in ocean salinity and temperature gradients. B. Landslides (into or under water), volcanic activity and meteorite impacts. C. Only prolonged monsoon rainfall over the ocean surface. D. Only anthropogenic seabed dredging near ports.

Answer: B. Explanation: VisionIAS names landslides, volcanic activity and meteorite impacts as additional tsunami triggers alongside undersea earthquakes. The remaining options name mechanisms (monsoon rainfall, seabed dredging, salinity/temperature change) the source does not identify as tsunami-generating triggers.

Q3. Which combination most directly determines how a tsunami affects a particular coast?

A. Only the population density of the nearest coastal district. B. Only the number of fishing boats at sea at the time of the event. C. Characteristics of the generating seismic event, distance from origin/magnitude, and bathymetry (ocean depth configuration). D. Only the calendar month in which the undersea earthquake occurs.

Answer: C. Explanation: The source ties tsunami effect to the generating event's characteristics, distance/magnitude from origin, and bathymetry. The remaining options (calendar month, district population density, fishing-boat count) name factors that do not determine wave generation or shoaling behaviour.

Q4. Which statement correctly separates a tsunami from a tide and a cyclone-driven storm surge?

A. A tsunami, a tide and a storm surge are three names for the same coastal-water phenomenon. B. A storm surge is caused by seabed displacement, exactly like a tsunami. C. A tide is generated by undersea earthquakes in the same way a tsunami is. D. A tsunami begins with sudden water displacement, unlike a tide (a gravitational cycle) or a storm surge (chiefly wind-and-pressure driven).

Answer: D. Explanation: Tides follow gravitational cycles and storm surges are wind-and-pressure driven, while a tsunami's generation begins with sudden water displacement. The remaining options each wrongly collapse these three distinct mechanisms into one.

Q5. Which statement accurately reports the document-period historical record of India's tsunami exposure?

A. More than 2,200 km of heavily populated shoreline is exposed, with 13 tsunamis recorded in the Indian Ocean region in the last 300 years, 3 of them in the Andaman and Nicobar region. B. Only the western coastline of India carries any tsunami exposure; the eastern coastline is unaffected. C. India's coastline has recorded exactly one tsunami in its documented history. D. Tsunami exposure in India is confined entirely to Lakshadweep and excludes the mainland coast.

Answer: A. Explanation: The source records more than 2,200 km of exposed shoreline and 13 tsunamis in 300 years across the Indian Ocean region, 3 in the Andaman and Nicobar region -- a document-period historical figure, not a verified present-day statistic; cite current INCOIS/GSI data before treating either number as today's fact. The remaining options each misstate the geographic spread or historical count given in the source.

Q6. Which event is recognised as the most destructive tsunami to have affected India?

A. A tsunami generated by the 1934 Bihar-Nepal earthquake. B. The 26 December 2004 Indian Ocean Tsunami. C. A tsunami triggered by the 2001 Bhuj earthquake. D. A tsunami following the 1993 Latur earthquake.

Answer: B. Explanation: VisionIAS names the 26 December 2004 Indian Ocean Tsunami as the most destructive tsunami to have hit India. The remaining options name inland or non-tsunamigenic earthquake events the source does not describe as tsunami sources.

Q7. Which damage pattern was observed during the 2004 Indian Ocean tsunami?

A. Only artificial sea walls, and no natural feature, reduced tsunami damage in 2004. B. Sand-dune mining had no bearing on the extent of damage recorded in 2004. C. Maximum damage occurred in low-lying, densely populated coastal areas, while mangroves, forests, sand dunes and coastal cliffs provided the best natural barriers; damage was heavy where dunes were mined. D. Damage was uniform across every coastal district regardless of elevation, population density or natural cover.

Answer: C. Explanation: The source ties maximum 2004 damage to low-lying, densely populated areas and credits mangroves/forests/dunes/cliffs as the best natural barriers, with heavy damage where dunes were mined. The remaining options each contradict this recorded damage pattern.

Q8. Which institutions participated in establishing India's Tsunami Early Warning System?

A. Only the Ministry of Home Affairs, acting independently of any scientific department. B. Only the Reserve Bank of India, for disaster-finance purposes. C. Only state Public Works Departments, without any central scientific institution. D. The Department of Space, the Department of Science and Technology, and CSIR.

Answer: D. Explanation: The source states the warning system was established with the Department of Space, DST and CSIR. The remaining options name institutions the source does not credit with establishing this warning system.

Q9. Which statement about the Indian Tsunami Early Warning Centre is accurate?

A. ITEWC, at INCOIS, Hyderabad, has been operational since 15 October 2007, detects tsunamigenic earthquakes in about 10 minutes and disseminates advisories in about 20 minutes. B. ITEWC was established in Chennai in 1999 and issues advisories only once every 24 hours. C. ITEWC detects and disseminates advisories in the same roughly 10-minute window, with no distinction between the two stages. D. ITEWC is a state-government body under Tamil Nadu's disaster-management department.

Answer: A. Explanation: ITEWC, housed at INCOIS Hyderabad, has been operational since 15 October 2007, with about 10 minutes for detection and about 20 minutes for advisory dissemination. The remaining options misstate its location/founding date, collapse the detection-dissemination gap, or misstate its institutional responsibility.

Q10. Under which regional framework does ITEWC operate as an approved Tsunami Service Provider?

A. The Association of Southeast Asian Nations' disaster-management committee acting alone. B. The UNESCO Intergovernmental Oceanographic Commission's Indian Ocean Tsunami Warning and Mitigation System (IOTWMS). C. The World Meteorological Organization's Tropical Cyclone Programme. D. The United Nations Framework Convention on Climate Change's adaptation fund mechanism.

Answer: B. Explanation: ITEWC is an approved Tsunami Service Provider under UNESCO-IOC's IOTWMS, issuing advisories for India and the wider region. The remaining options name unrelated international frameworks that do not designate ITEWC as a service provider.

Q11. Which model was jointly customised by INCOIS and ICMAM for tsunami surge and inundation assessment?

A. The Standard Penetration Test model used in soil mechanics. B. The Palmer Drought Severity Index model. C. The TUNAMI-N2 model. D. The Köppen climate classification model.

Answer: C. Explanation: INCOIS and the Integrated Coastal and Marine Area Management (ICMAM) jointly customised the TUNAMI-N2 model for surge/inundation prediction across earthquake scenarios. The remaining options name models used for unrelated purposes: climate classification, soil testing and drought measurement, none of which is a tsunami-surge model.

Q12. How does CRZ Notification 2019 support tsunami-safe coastal development?

A. Through one single, universal minimum-distance rule applied identically to every Indian coastal state. B. Through a single elevation threshold that overrides all category-specific coastal-zone distinctions. C. By exempting all coastal construction from any regulation once a bio-shield is planted nearby. D. Through category-specific controls (CRZ-I to CRZ-IV), each with distinct no-development/regulated-development zones, combined with tsunami inundation and hazard mapping.

Answer: D. Explanation: CRZ Notification, 2019 uses category-specific controls (CRZ-I to CRZ-IV) combined with hazard mapping, not one universal rule. The remaining options each invent a single blanket rule the notification does not use.

Q13. Which description best defines a coastal bio-shield?

A. A narrow, thickly planted strip of land along the coastline, developed as a coastal-zone disaster-management sanctuary with public-awareness spaces. B. A concrete sea wall reinforced with steel embedded along the entire coastline. C. A satellite-based early-warning transmitter installed on offshore buoys. D. A government cash-transfer scheme paid to fisherfolk after a tsunami.

Answer: A. Explanation: A bio-shield is defined as a narrow, thickly planted coastal strip functioning as a disaster-management sanctuary with awareness spaces. The remaining options describe structural, technological and financial measures, none of which match the source's bio-shield definition.

Q14. Which option combines structural and ecosystem-based coastal-hazard measures?

A. Only satellite monitoring, with no on-ground physical mitigation measure named. B. Cyclone shelters, submerged sand barriers/dykes, sand dunes with sea-weed planting, mangrove/coastal-forest plantation, sea walls, coral reefs and breakwaters. C. Only concrete sea walls, with no role assigned to mangroves, dunes or coral reefs. D. Only afforestation programmes, with no structural measures named at all.

Answer: B. Explanation: The source names both structural measures (shelters, barriers/dykes, sea walls, breakwaters) and ecosystem-based measures (dunes with sea-weed planting, mangrove/forest plantation, coral reefs) together. The remaining options each drop one category the source explicitly retains alongside the other.

Q15. What bathymetry-related role does the Indian Naval Hydrographic Department perform for tsunami preparedness?

A. It operates the Bottom Pressure Recorders that detect open-ocean wave propagation. B. It licenses engineers and masons for tsunami-resistant coastal construction. C. It regularly supplies bathymetry information for inundation maps, supporting NDMA's vulnerability/risk mapping using coastal land-use data. D. It issues the final local evacuation order for every coastal district during a tsunami alert.

Answer: C. Explanation: INHD regularly supplies bathymetry data for inundation maps feeding NDMA's coastal vulnerability/risk mapping. The remaining options describe local-authority evacuation powers, the seismic/BPR detection subsystem and a licensing function, none of which the source assigns to INHD.

Q16. Which communities were India's first UNESCO-IOC Tsunami Ready recognitions, and when?

A. Chennai and Puri, recognised as fully Tsunami Ready states in 2007. B. Port Blair and Kavaratti, recognised as national Tsunami Ready certifications in 2004. C. Visakhapatnam and Kochi, recognised in 2019 as India's only Tsunami Ready cities. D. Venkatraipur (Ganjam) and Noliasahi (Jagatsinghpur) in Odisha, recognised on 7 August 2020.

Answer: D. Explanation: Venkatraipur (Ganjam) and Noliasahi (Jagatsinghpur), Odisha, were recognised on 7 August 2020 as India's first Tsunami Ready communities. The remaining options name places, dates or a national-certification framing the source does not support; recognition is community-level, not statewide or national.

PYQS AND ANSWER PRACTICE

Verified PYQ coverage

The 2025 GS-I tsunami-formation question is used for the disaster-management application of detection, warning and evacuation. The 2023 GS-I coastline question is addressed only for coastal hazard preparedness, and the 2024 GS-III resilience question is applied through a tsunami-specific lens. The 2025 seawater-intrusion question is adjacent coastal-water content, not a tsunami question. No verified 2026 Prelims or Mains question routes directly to this topic.

PYQ DEMAND CARD 1 - 2025 GS-I

Demand: Explain tsunami formation and consequences. (Explain · 10 marks · about 150 words)

Status: This card focuses on the disaster-management application; the physical-process explanation is a GS-I Geography demand.

Model answer (145 words): A tsunami is a series of long waves generated by sudden displacement of a large water volume, most commonly vertical seabed movement in an undersea earthquake, but also landslides, volcanic activity or meteorite impacts. Generation is not impact: an undersea earthquake does not automatically produce a destructive tsunami, since trigger distance, magnitude, bathymetry and coastal geometry govern whether and how waves affect a coast. In deep water the waves travel fast with low amplitude; entering shallow water, they shoal, slow and grow in height, producing run-up, inundation and a damaging return flow. Consequences include coastal flooding, structural damage, loss of life and livelihoods, and disruption to ports, fisheries and tourism. India's Tsunami Early Warning Centre (ITEWC) at INCOIS integrates seismic analysis with sea-level observation to confirm generation and update threat assessments, but the local evacuation all-clear remains a competent-authority decision, not a warning-centre output alone.

Examiner note: Named evidence (seabed displacement, shoaling/run-up, ITEWC's detection-to-advisory chain) does the evidentiary work; do not invent an unsourced wave-height, speed or casualty figure.

Answer plan: trigger set and definition -> generation-is-not-impact distinction (bathymetry/distance) -> deep-water-to-shoaling mechanism -> named consequences -> one warning-chain closing line.

Improvement pointer: name the 2004 event only as a one-clause mechanism illustration, not a casualty recital, to keep the answer analytical rather than descriptive.

PYQ DEMAND CARD 2 - 2023 GS-I

Demand: Comment on the resource potential of India's coastline and highlight hazard preparedness. (Comment and highlight · 15 marks · about 250 words)

Status: This card addresses only the tsunami and coastal-preparedness dimensions of the wider coastline demand.

Model answer (234 words): India's coastline exceeds 2,200 km across the East and West (a document-period figure), carrying genuine resource potential in fisheries, ports, shipping, tourism and offshore energy, alongside recurring hazard exposure: 13 tsunamis on historical record over the last 300 years, three in the Andaman and Nicobar region, besides cyclones, storm surge and coastal erosion. Realising this potential safely depends on a linked preparedness chain rather than any single measure. Detection and warning: ITEWC at INCOIS integrates seismic analysis with open-ocean Bottom Pressure Recorders and coastal tide gauges, issuing staged bulletins in roughly 10 to 20 minutes as an approved UNESCO-IOC Tsunami Service Provider under IOTWMS. Mapping and regulation: the TUNAMI-N2 model and Naval Hydrographic Department bathymetry feed inundation maps, while the CRZ Notification, 2019's category-specific CRZ-I to CRZ-IV controls regulate risk-sensitive coastal development rather than imposing one universal setback. Protection: structural measures (sea walls, breakwaters, submerged sand barriers and dykes) work alongside ecosystem-based buffers (mangroves, coastal forests, sand dunes with sea-weed planting and bio-shields), since the 2004 tsunami showed heavier damage precisely where dunes were mined. Response and recovery: community first response, SHG/NGO/CBO-supported search and rescue, and the Naval Hydrographic Department's seven-ship 2004 deployment illustrate multi-agency capacity. Last-mile safety closes the chain: accessible evacuation routes, drills and UNESCO-IOC 'Tsunami Ready' community recognition, beginning with Venkatraipur and Noliasahi in Odisha in 2020, convert coastline potential into genuinely resilient use rather than unmanaged exposure.

Examiner note: Keep resource potential (fisheries/ports/tourism) and hazard preparedness (ITEWC, CRZ, structural/ecosystem buffers, Tsunami Ready) as two linked but distinct halves of the answer.

Answer plan: open on resource potential -> pivot explicitly to hazard exposure -> sequence the preparedness chain (detection -> mapping -> protection -> response -> last-mile) -> close by linking potential to preparedness.

Improvement pointer: name at least one ecosystem-based measure (e.g. mangrove plantation) alongside a structural one, since balanced-mitigation knowledge is specifically rewarded here.

PYQ DEMAND CARD 3 - 2024 GS-III

Demand: Describe the elements that determine disaster resilience.

Directive and limit: Describe · 15 marks · about 250 words.

Status: This card applies the general disaster-resilience framework specifically to tsunami warning, land use, buffers, evacuation and community readiness.

Model answer (212 words): Disaster resilience is the capacity of a system or community to anticipate, absorb, respond to and recover from a hazard while preserving essential functions. In a tsunami context, six linked elements determine this capacity. First, scientific capability: ITEWC at INCOIS must combine seismic analysis, models, Bottom Pressure Recorders and tide gauges to issue staged advisories. Second, risk knowledge: TUNAMI-N2 scenarios, bathymetry and local topography should guide inundation maps, evacuation zones and critical-facility siting. Third, risk-sensitive development: CRZ 2019 categories and local land-use controls must limit new exposure. Fourth, layered protection: engineered shelters, retrofits or breakwaters should be combined with dunes, mangroves, reefs and other ecosystem buffers; the 2004 damage pattern where dunes were mined shows why degradation weakens resilience. Fifth, inclusive last-mile capacity: multilingual warnings, accessible horizontal or designated vertical evacuation, repeated drills and support for fishers, tourists, older persons and persons with disabilities convert an advisory into safe action. Sixth, institutional continuity: UNESCO-IOC Tsunami Ready recognition, first achieved in India by Venkatraipur and Noliasahi in 2020, must be renewed so rare-hazard readiness does not decay. Recovery should then restore ports, fisheries, housing and services through Build Back Better. Thus resilience is produced by a maintained chain from knowledge and prevention to local action and adaptive recovery, not by one warning technology.

Answer plan: define resilience -> organise the six tsunami-specific elements -> attach 2004 and Tsunami Ready evidence -> conclude with the maintained-chain test.

Improvement pointer: keep the answer disaster-specific while showing that scientific, social, institutional, ecological and recovery capacities operate together.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish tsunami generation from tides and cyclone-driven storm surge. Answer in about 150 words.

Model answer (138 words): A tsunami is not an astronomical tide and not a cyclone-driven storm surge. Tides follow predictable gravitational cycles of the moon and sun; a storm surge is chiefly wind-and-pressure driven, building up during a cyclone's approach. A tsunami's generation instead begins with sudden displacement of a large water volume, most commonly vertical seabed movement during an undersea earthquake, though landslides, volcanic activity and meteorite impacts are alternative triggers. This distinction matters operationally: tide tables and cyclone-track forecasts cannot substitute for seismic and sea-level monitoring, since an undersea earthquake does not automatically produce a destructive tsunami -- source mechanism, depth, displacement, distance, bathymetry and coastal geometry determine whether and how waves actually affect a coast. Conflating the three phenomena would misdirect both warning systems and public understanding, since each requires a distinct detection method, lead time and institutional response chain.

Answer plan: negative definition first (not a tide, not a storm surge) -> driver-based distinction in prose -> tie back to operational consequence (wrong monitoring tool used).

Improvement pointer: name the specific instrument each phenomenon needs (tide tables; cyclone-track forecasts; seismic/sea-level monitoring) in one explicit sentence for a sharper close.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the detection, forecast, warning and local all-clear chain for tsunamis. Answer in about 150 words.

Model answer (144 words): Tsunami detection, forecast, warning and the local all-clear are four distinct stages, not one collapsed step. Detection begins with seismic analysis identifying an undersea earthquake's parameters, cross-checked by open-ocean Bottom Pressure Recorders and coastal tide gauges confirming actual wave generation. India's Tsunami Early Warning Centre (ITEWC) at INCOIS, Hyderabad, operational since 15 October 2007, performs this detection in roughly 10 minutes. Forecast follows, using the TUNAMI-N2 model (customised by ICMAM and INCOIS) to estimate likely surge and inundation for the confirmed scenario. ITEWC then issues a staged bulletin -- threat category and advisory -- within roughly 20 minutes, functioning as an approved UNESCO-IOC Tsunami Service Provider under the IOTWMS for India and the region. The local all-clear, however, is not an ITEWC output: competent local authorities must weigh local conditions before declaring safety, since a regional advisory is evidence for a decision, not the decision itself.

Answer plan: state the four-stage claim upfront -> walk each stage in strict order -> close on the local-authority boundary.

Improvement pointer: cite the two time figures (roughly 10 minutes / 20 minutes) explicitly and distinctly, since examiners specifically reward not collapsing them into one number.

ORIGINAL MAINS 3 - 15 MARKS

Question: Discuss how India combines structural and ecosystem-based measures for coastal-hazard mitigation. Answer in about 250 words.

Model answer (233 words): Coastal-hazard mitigation combines structural and ecosystem-based measures, since the 2004 Indian Ocean tsunami showed neither works alone. Structural measures include cyclone shelters, submerged sand barriers and dykes, location-specific sea walls and breakwaters, plus identification of vulnerable structures for retrofitting; each targets a specific stretch of coast, not one universal design. Ecosystem-based measures include sand dunes planted with sea-weed, mangrove and coastal-forest plantation, bio-shields, and coral reefs -- natural buffers, some doubling as disaster-management sanctuaries with public-awareness spaces. The 2004 event showed mangroves, forests, dunes and cliffs provided the best natural barriers, while heavy damage occurred specifically where dunes had been mined, proving that removing an ecosystem buffer itself raises risk, not a neutral, cost-free choice. The CRZ Notification, 2019's category-specific CRZ-I to CRZ-IV controls, combined with tsunami inundation/hazard mapping and MHA's model techno-legal framework for zoning, planning, design and construction, give this combination a regulatory backbone. Treating either category as sufficient alone -- engineered walls without

buffers, or buffers without retrofitted structures -- would leave a coast only partially protected against a low-frequency, high-consequence hazard. The M.S. Swaminathan Research Foundation, Pondicherry, functions as a local-knowledge centre linking community awareness to engineered design, while NDMA's vulnerability mapping keeps the structural side updated. A balanced answer names both categories together, cites the 2004 dune-mining evidence for why ecosystem removal is not cost-free, and roots the CRZ-plus-MHA backbone as the mechanism binding both into one coherent system.

Answer plan: open with the complementary-categories thesis -> list structural measures -> list ecosystem-based measures (including coral reefs) -> cite 2004 dune-mining evidence -> close on the regulatory backbone (CRZ+MHA).

Improvement pointer: state explicitly that ecosystem-based measures are not secondary to structural ones -- examiners reward this named distinction over a flat list of measures.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse the institutional architecture and limits of India's tsunami detection-to-advisory system. Answer in about 250 words.

Model answer (221 words): India's tsunami detection-to-advisory architecture is genuinely capable but structurally multi-agency and resource-dependent. The Indian Tsunami Early Warning System was established with the Department of Space, the Department of Science and Technology and CSIR; its seismic subsystem draws real-time data from IMD's national seismic network, together with international networks, while Bottom Pressure Recorders and tide gauges confirm open-ocean wave propagation. ITEWC at INCOIS, Hyderabad, operational since 15 October 2007, integrates this evidence to issue bulletins in roughly 10 minutes for detection and 20 minutes for advisory dissemination, as an approved UNESCO-IOC Tsunami Service Provider under IOTWMS for India and the region. The Indian Naval Hydrographic Department regularly supplies bathymetry for inundation mapping, and famously deployed seven survey ships after 2004 to reopen sea lines of communication and re-chart the seabed, while the TUNAMI-N2 model, customised by ICMAM and INCOIS, converts this evidence into evacuation, settlement-avoidance and structural-design decisions. This capability's real limits are institutional rather than purely technical: sustaining seismic-station upkeep, periodic bathymetric resurveying and drilled community response between rare events is harder to maintain politically and financially than for annually recurring hazards such as cyclones, so continued institutional discipline -- not one-time technology deployment -- determines whether fast detection actually converts into last-mile local safety, and the local all-clear decision must still rest with competent local authorities rather than the warning centre itself.

Answer plan: name the multi-agency origin -> trace the seismic-to-advisory chain with the two time figures -> name the mapping/model layer -> close on institutional (not technical) limits.

Improvement pointer: end with an explicit sentence assigning the all-clear decision to local authorities, not ITEWC, to close the institutional-boundary argument being tested.

ORIGINAL MAINS 5 - 20 MARKS

Question: Critically examine UNESCO-IOC 'Tsunami Ready' recognition as an indicator of India's coastal resilience. Answer in about 250 words.

Model answer (247 words): UNESCO-IOC's 'Tsunami Ready' recognition is a community-level, indicator-based preparedness programme, not a national certification, and this distinction should anchor any evaluation of India's coastal resilience. Venkatraipur (Ganjam) and Noliasahi (Jagatsinghpur) in Odisha became India's first recognised communities on 7 August 2020, having demonstrated maintained warning-relay, mapped evacuation-route and drill indicators specific to their locality, not a statewide or national outcome. This granular design suits a low-frequency, high-consequence hazard: a national-level claim would obscure the last-mile reality that safety depends on trusted local relay, accessible evacuation infrastructure, inclusive assistance for fisherfolk, harbour workers and tourists, and repeated drills -- none guaranteed by ITEWC's 10-to-20-minute detection chain alone. The risk is that recognising two communities is quietly generalised into an unverified claim of coast-wide readiness, since institutional attention to rare hazards spikes after a disaster and decays before the next one. A sound evaluation should instead track how many more communities are recognised over time, whether recognition is periodically renewed rather than a one-time badge, whether indicators match observed drill performance rather than the ITEWC bulletin chain alone, and whether structural retrofitting, mangrove/dune protection and CRZ-linked land-use discipline advance alongside the recognition

programme. The benchmark indicators -- hazard mapping, warning dissemination, public education, periodic drills -- work only where local governance sustains them between disasters; a 2020 badge says nothing about whether relay equipment still functions or new arrivals have been briefed since. Genuine coastal resilience is best read as a renewable, continuously maintained process, not a fixed, one-time national status.

Answer plan: state the community-vs-national boundary as the thesis -> give the named evidence (Venkatraipur/Noliasahi, 2020) -> argue why granularity suits a low-frequency hazard -> name the generalisation risk -> close with renewal-based criteria for genuine resilience.

Improvement pointer: add one explicit recommendation (e.g. publish an annual renewed-recognition count) to convert the critique into an actionable improvement, which 20-mark 'critically examine' answers are specifically expected to supply.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an exam-safe tsunami-resilience framework covering detection, land use, protection, evacuation and recovery. Answer in about 250 words.

Model answer (245 words): An exam-safe tsunami-resilience framework separates five linked layers, not one blurred claim. First, detection: ITEWC/INCOIS's seismic-plus-sea-level chain (Department of Space, DST, CSIR; Bottom Pressure Recorders) delivers roughly 10-minute detection and 20-minute advisory dissemination as an approved UNESCO-IOC Tsunami Service Provider. Second, mapping and land use: TUNAMI-N2 model outputs and Naval Hydrographic Department bathymetry feed inundation maps, matched to the CRZ Notification, 2019's category-specific CRZ-I to CRZ-IV controls and MHA's techno-legal framework. Third, mixed protection: structural measures (sea walls, breakwaters, sand barriers/dykes, retrofitted structures) alongside ecosystem-based buffers (mangroves, coastal forests, dunes, bio-shields, coral reefs), since 2004 showed heavier damage where dunes were mined. Fourth, evacuation readiness: accessible warning relay, mapped routes, community first response with SHG/NGO/CBO support, drills and UNESCO-IOC 'Tsunami Ready' recognition, renewed rather than permanent. Fifth, recovery: restoring ports, fisheries and tourism via Build Back Better siting, supported by inflatable boats, helicopters and search-rescue equipment immediately post-event. Local all-clear authority stays with competent local administration throughout, since no single layer alone proves last-mile safety. A sixth, governance thread cuts across all five: State and District Disaster Management Authorities under the Disaster Management Act, 2005 must translate NDMA's guidelines into locally funded, evaluated and accountable action, since a rare hazard otherwise suffers readiness-decay as budgets and drills quietly lapse. A full-marks answer sequences these six threads, cites 2004 as ITEWC's institutional origin point, and closes by naming continuous inter-agency coordination -- not one technology or badge -- as the true measure of resilience.

Answer plan: name the five layers upfront as an explicit list -> develop each layer in one tight sentence with named evidence -> add the cross-cutting governance thread -> close by naming the single true measure of resilience.

Improvement pointer: state which single layer would fail first without institutional discipline (readiness-decay) to show evaluative depth beyond a descriptive five-layer list.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Low-frequency hazard and readiness decay

Tsunamis are rare enough for public memory, budgets, sensor maintenance, signage and drills to erode between events, yet consequences can be catastrophic. The policy problem is therefore not only building capability after a disaster but institutionalising renewal before the next one.

Data-quality and infrastructure limits

Fast earthquake detection and accurate local inundation prediction are different capacities. Near-shore bathymetry, topography, land-cover data and functioning observation platforms determine forecast precision. Redundancy and tamper-resistant sensors matter because ocean platforms may fail or be damaged.

Structural and ecosystem trade-offs

Engineered barriers offer designed performance but face overtopping, scour, maintenance and ecological costs. Dunes, mangroves, reefs and forests offer friction and co-benefits but depend on width, health, location and hazard scale. The defensible recommendation is a mapped, location-specific portfolio rather than faith in one barrier.

Advanced answer framework

1. Name the low-frequency/readiness-decay problem.
2. Separate detection speed from local forecast precision.
3. Trace advisory authority to last-mile evacuation.
4. Combine CRZ-linked land-use discipline with engineered and ecosystem protection.
5. Include ports, fisheries, tourists, islands and persons requiring assisted evacuation.
6. Evaluate outcomes through drills, route usability, renewed recognition and recovery performance.

CONSOLIDATED REGISTER NOTES

Core mechanism

```
sudden water displacement -> long-wave propagation -> shoaling
-> run-up / inundation / return flow -> coast-specific consequences
```

Warning chain

```
seismic analysis -> model-based threat -> BPR/tide-gauge confirmation
-> ITEWC bulletin -> State/district/local action -> authorised all-clear
```

Essential facts

Item	Revision line
ITEWC	At INCOIS, Hyderabad; operational since 15 October 2007.
Timing	About 10 minutes for earthquake detection; about 20 minutes for advisory dissemination.
Regional role	Approved UNESCO-IOC Tsunami Service Provider under IOTWMS.
Tsunami Ready	Community-level, renewable recognition; Venkatraipur and Noliasahi were India's first recognised communities on 7 August 2020.
Modelling	TUNAMI-N2 plus bathymetry/topography supports inundation and evacuation planning.
Regulation	CRZ 2019 categories and hazard maps support risk-sensitive siting; no universal safe distance.

Indian source geography and the 2004 evidence base

- **Eastern source:** Andaman-Sumatra subduction zone; **western source:** Makran subduction zone.
- **Trigger range:** vertical seabed movement, submarine landslide, volcanic collapse and rare meteorite impact.
- **26 December 2004:** use as the institutional turning point for India's warning architecture.
- **Damage lesson:** low-lying, densely populated coasts suffered heavily; mangroves, forests, dunes and cliffs reduced damage under specific site conditions, while damage was heavier where dunes were mined.
- **Document-period exposure figures:** more than 2,200 km of heavily populated shoreline and 13 Indian Ocean tsunamis in 300 years, including three in the Andaman and Nicobar region. Verify against a current INCOIS/GSI source before presenting these as present-day counts.

Institutional and technical evidence

- The Indian Tsunami Early Warning System was established with the **Department of Space, Department of Science and Technology and CSIR**.
- **ITEWC at INCOIS** integrates seismic analysis, **Bottom Pressure Recorders**, tide gauges and modelling.
- **TUNAMI-N2**, customised by ICMAM and INCOIS, supports surge and inundation scenarios.
- The **Indian Naval Hydrographic Department** supplies bathymetry and deployed seven survey ships after the 2004 disaster for aid, sea-lane reopening and seabed re-charting.
- The **M.S. Swaminathan Research Foundation, Pondicherry**, is a local-knowledge-centre example linking community awareness with coastal preparedness.
- A **bio-shield** is a thickly planted coastal strip; it belongs in a layered portfolio with dunes, mangroves, reefs, shelters, retrofits, walls and breakwaters.

Response, recovery and Build Back Better

- Immediate priorities: evacuation, accounting, search and rescue, health, safe water, sanitation, shelter and protection.
- Restore ports, fisheries, tourism, utilities, housing and ecosystems while updating maps, routes and construction practice.
- **Build Back Better** means safer siting and design without severing fisherfolk or other coastal communities from livelihood access and rights.

High-yield traps

- A tsunami is not a tide or cyclone storm surge.
- Not every undersea earthquake is tsunamigenic.
- The first wave need not be the largest.
- Run-up, offshore height and inundation distance are different measures.
- Detection is not confirmation; advisory is not evacuation; ITEWC does not issue the local all-clear.
- Tsunami Ready is community recognition, not national certification.
- Mangroves, dunes or sea walls reduce risk but do not eliminate it.
- Tsunami genesis is geological, not climate-induced; sea-level rise may alter the coastal exposure baseline.

Mains answer spine

Opening: Tsunami risk is a low-frequency, high-consequence chain from seabed displacement to coast-specific impact.

Body: trigger -> propagation/shoaling -> Indian warning architecture -> mapping/CRZ -> layered protection -> inclusive evacuation -> livelihood-sensitive recovery.

Evaluation: test the full chain from technical advisory to safe last-mile action rather than praising one technology.

ASCII master flowchart

ASCII MASTER FLOW - PANEL 1/12: Generation and hazard firewall

```
EARTHQUAKE | LANDSLIDE | VOLCANIC ACTIVITY | METEORITE IMPACT
-> RAPID WATER DISPLACEMENT -> LONG TSUNAMI WAVES
TSUNAMI != TIDE (GRAVITY) != STORM SURGE (CYCLONE WIND/PRESSURE)
UNDERSEA EARTHQUAKE ALONE IS NOT SUFFICIENT: VERTICAL DISPLACEMENT MATTERS
```

ASCII MASTER FLOW - PANEL 2/12: Propagation to coastal impact

```
DEEP OCEAN -> FAST TRAVEL + LOW VISIBLE AMPLITUDE
SHALLOW WATER -> SLOWING + SHOALING + HEIGHT/CURRENT INCREASE
-> RUN-UP (VERTICAL) + INUNDATION (HORIZONTAL) + RETURN FLOW
FIRST WAVE NEED NOT BE LARGEST -> WAIT FOR AUTHORISED ALL-CLEAR
```

ASCII MASTER FLOW - PANEL 3/12: Indian source geography and exposure

```
EAST -> ANDAMAN-SUMATRA SUBDUCTION ZONE
WEST -> MAKRAN SUBDUCTION ZONE
26 DECEMBER 2004 -> CENTRAL INDIAN OCEAN DISASTER AND INSTITUTIONAL TURNING POINT
DOCUMENT-PERIOD FIGURES: >2,200 KM EXPOSED SHORELINE; 13 TSUNAMIS/300 YEARS, 3 IN ANDAMAN-NICOBAR
QUALIFY HISTORICAL FIGURES BEFORE USING THEM AS CURRENT COUNTS
```

ASCII MASTER FLOW - PANEL 4/12: Detection-confirmation-advisory rail

```
SEISMIC NETWORK -> SOURCE ANALYSIS -> TUNAMI-N2 SCENARIO
-> BOTTOM PRESSURE RECORDERS (OPEN OCEAN) + TIDE GAUGES (COAST)
-> OBSERVATION-INFORMED ITEWC BULLETIN / THREAT UPDATE
DETECTION != CONFIRMATION != ADVISORY != LOCAL EVACUATION
```

ASCII MASTER FLOW - PANEL 5/12: ITEWC authority and timing

```
ITEWC AT INCOIS, HYDERABAD -> OPERATIONAL SINCE 15 OCTOBER 2007
ABOUT 10 MINUTES -> EARTHQUAKE DETECTION
ABOUT 20 MINUTES -> ADVISORY DISSEMINATION
UNESCO-IOC IOTWMS -> REGIONAL TSUNAMI SERVICE FRAMEWORK
STATE / DISTRICT / LOCAL AUTHORITY -> EVACUATION AND ALL-CLEAR
```

ASCII MASTER FLOW - PANEL 6/12: Modelling and bathymetry

```
SOURCE SCENARIO + BATHYMETRY + TOPOGRAPHY + LAND COVER
-> TUNAMI-N2 -> TRAVEL TIME / RUN-UP / DEPTH / VELOCITY / INUNDATION
INDIAN NAVAL HYDROGRAPHIC DEPARTMENT -> BATHYMETRY AND 2004 RE-CHARTING
MODEL BOUNDARY = SCENARIO WITH UNCERTAINTY, NOT AN EXACT FUTURE LIMIT
```

ASCII MASTER FLOW - PANEL 7/12: Evacuation decision

```
STRONG/LONG SHAKING OR ABNORMAL SEA CHANGE -> SELF-EVACUATE
HORIZONTAL -> INLAND / HIGHER GROUND WHERE TIME ALLOWS
VERTICAL -> ENGINEERED, ACCESSIBLE, OFFICIALLY DESIGNATED REFUGE
MAP -> SIGNED ROUTE -> ASSEMBLY -> ACCOUNTABILITY -> DRILL -> REVISION
```

ASCII MASTER FLOW - PANEL 8/12: CRZ and layered coastal protection

```
CRZ NOTIFICATION 2019 -> CATEGORY-SPECIFIC CRZ-I TO CRZ-IV CONTROLS
ENGINEERED -> SHELTERS / RETROFITS / SEA WALLS / BREAKWATERS
ECOSYSTEM -> DUNES / MANGROVES / FORESTS / REEFS / WETLANDS / BIO-SHIELD
2004 EVIDENCE -> HEAVIER DAMAGE WHERE DUNES WERE MINED
NO UNIVERSAL SETBACK, SEA WALL OR ECOSYSTEM BUFFER ELIMINATES RISK
```

ASCII MASTER FLOW - PANEL 9/12: Coastal economy and inclusive warning

```
PORTS -> SHUTDOWN / HAZARDOUS CARGO / CHANNEL REOPENING
FISHERIES -> AT-SEA WARNING / BOATS / LANDING CENTRES / MARKETS
TOURISM -> MULTILINGUAL SIGNAGE / UNFAMILIAR ROUTES
ISLANDS + OLDER PERSONS + DISABILITY + PATIENTS -> ASSISTED EVACUATION
CONTINUITY PLAN MUST MATCH USER, SECTOR AND MOBILITY
```

ASCII MASTER FLOW - PANEL 10/12: Tsunami Ready and readiness decay

```
VENKATRAIPUR (GANJAM) + NOLIASAHI (JAGATSINGHPUR), ODISHA
-> INDIA'S FIRST UNESCO-IOC TSUNAMI READY COMMUNITIES, 7 AUGUST 2020
MAP -> EDUCATE -> WARN -> DRILL -> EVALUATE INDICATORS -> RENEW
COMMUNITY-LEVEL RENEWABLE RECOGNITION != NATIONAL CERTIFICATION
COUNTER MEMORY LOSS, STAFF CHANGE, DEGRADED SIGNAGE AND ROUTE CHANGE
```

ASCII MASTER FLOW - PANEL 11/12: Response, recovery and named evidence

```
EVACUATION -> SEARCH/RESCUE -> HEALTH/WATER/SANITATION/SHELTER
INDIAN NAVAL HYDROGRAPHIC DEPARTMENT: 7 SURVEY SHIPS AFTER 2004
-> AID + SEA-LANE REOPENING + SEABED RE-CHARTING
PORTS / FISHERIES / TOURISM / HOUSING / ECOSYSTEMS -> BUILD BACK BETTER
SAFER SITING MUST NOT DESTROY LIVELIHOOD ACCESS OR COASTAL RIGHTS
```

ASCII MASTER FLOW - PANEL 12/12: UPSC answer spine and evidence limits

```
DEFINE DISPLACEMENT -> TRACE PROPAGATION / SHOALING / CONSEQUENCE
-> MAP ITEWC / BPR / TIDE GAUGE / TUNAMI-N2 / LOCAL AUTHORITY
-> APPLY CRZ 2019 + LAYERED PROTECTION + ACCESSIBLE EVACUATION
-> USE 2004 + TSUNAMI READY EVIDENCE -> RECOVERY AND MAINTAINED READINESS
EVIDENCE LIMIT: ATTRIBUTE DATES, COUNTS, LEAD TIMES AND DAMAGE CLAIMS
OUTCOME TEST: A FAST ADVISORY IS AN INPUT; SAFE LAST-MILE ACTION IS THE RESULT
```